Chapter 1

1.1

$$\vec{b} = \hat{x} + \hat{y}$$
$$\vec{c} = \hat{x} + \hat{z}$$

$$\vec{b} + \vec{c} = 2\hat{x} + \hat{y} + \hat{z}$$

$$5\vec{b} + 2\vec{c} = 7\hat{x} + 5\hat{y} + 2\hat{z}$$

$$\vec{b} \cdot \vec{c} = 1$$

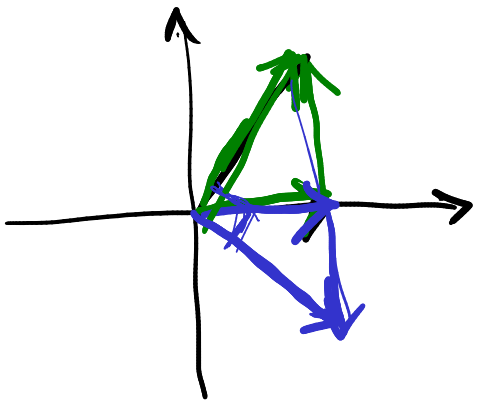
$$\vec{b} \times \vec{c} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = 1\hat{x} - (1)\hat{y} - 1\hat{z} = \hat{x} - \hat{y} - \hat{z}$$

1.6

$$\vec{b} = \hat{x} + s\hat{y}$$
$$\vec{c} = \hat{x} - s\hat{y}$$

$$\vec{b} \cdot \vec{c} = 1 - s^2 = 0$$

$$s = 1$$



1.24

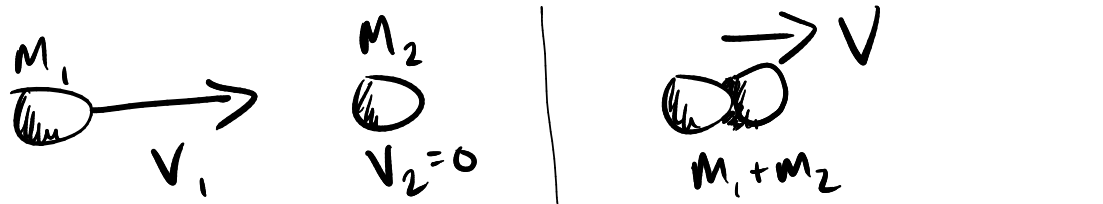
$$\frac{df}{dt} = kf$$

$$\int \frac{df}{f} = \int k\,dt$$

$$\ln f = kt + C$$

$$f = e^{t+C} = Ae^{kt}$$

1.30

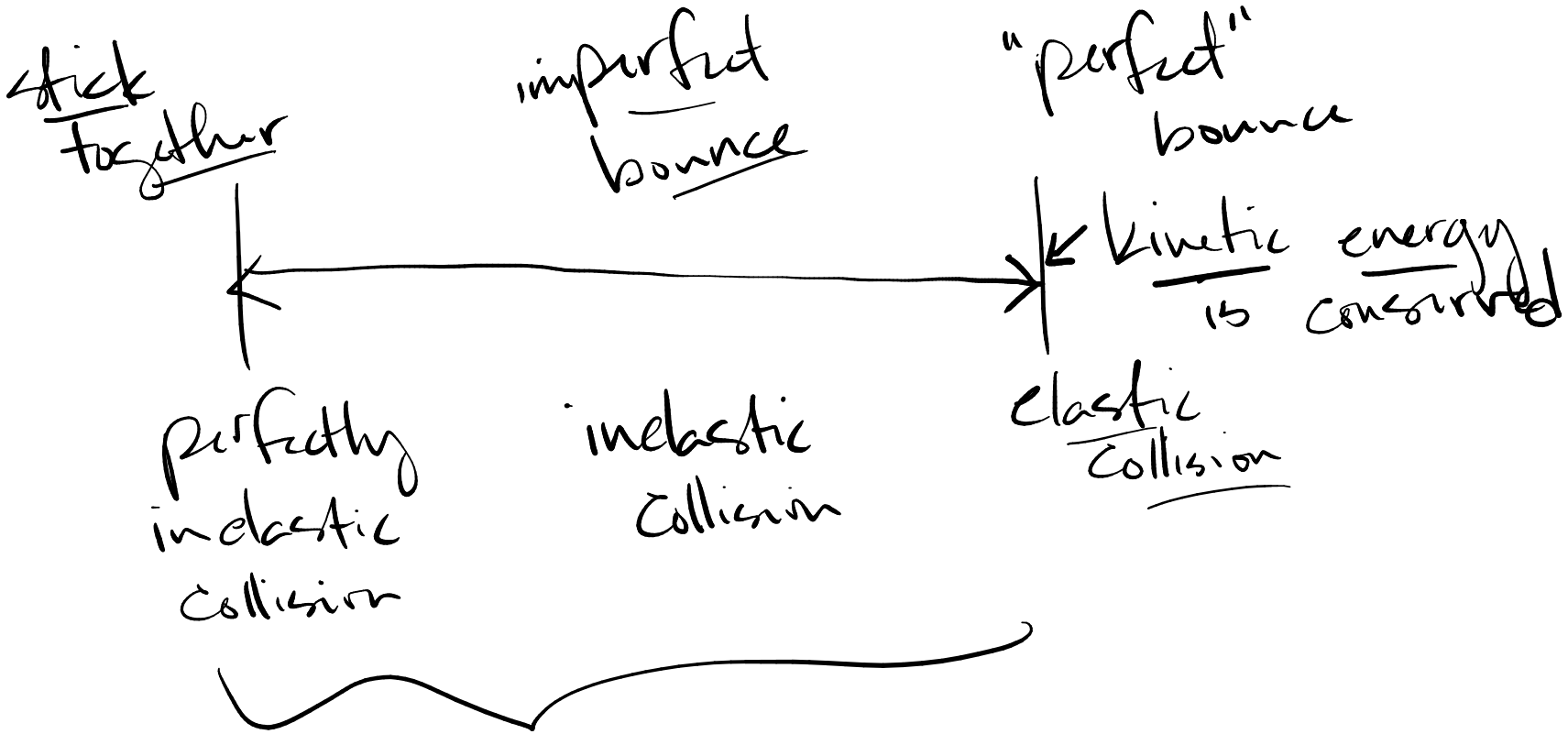


$$\vec{p}_i = m_1 v_1 + m_2 v_2 \quad (v_2 = 0)$$

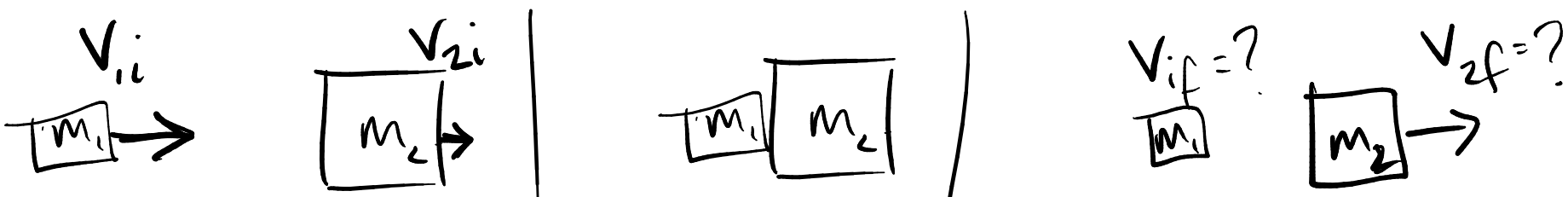
$$p_f = (m_1 + m_2)v$$

$$m_1 v_1 = (m_1 + m_2)v$$

$$v' \to v = \frac{m_1 v_1}{m_1 + m_2}$$


stick together
imperfect bounce
"perfect" bounce
kinetic energy is conserved
perfectly inelastic collision
inelastic collision
elastic collision
kinetic energy is not conserved
deformation, heat, sound

Elastic collisions



conservation of momentum

$$\vec{p}_i = m_1 v_{1i} + m_2 v_{2i} = \underbrace{m_1 v_{1f}}_{?} + \underbrace{m_2 v_{2f}}_{?} = \vec{p}_f$$

conservation of kinetic energy

$$\frac{1}{2} m_1 v_{1i}^2 + \frac{1}{2} m_2 v_{2i}^2 = \frac{1}{2} m_1 v_{1f}^2 + \frac{1}{2} m_2 v_{2f}^2$$

$v_{1f} = ?$

$v_{2f} = ?$

$\left(m_1, m_2, v_{1i}, v_{2i} \right)$